

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. The kinetic state is the closed graph of interior transport and compatible boundary trace, so collision-time atoms are legitimate without independent arbitrary traces. A monotone hard-sphere Jacobi form prevents later dispersing collisions from focusing the youngest-fork transverse plane. This gives depth-independent recollision power, source-uniform sewing, and balance-preserving path-law regularization.

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1. ROUND-EIGHT RECONSTRUCTION: COMPATIBLE KINETIC TRACES AND MONOTONE FORK JACOBI FIELDS

Boundary flux is not an independent coordinate. We define the hierarchy as a closed graph of transport states satisfying the Green identity and collision compatibilities. The recollision estimate is proved in the physical Jacobi metric of a dispersing hard-sphere flow; no coordinate reset is used to erase singular values.

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1.1. The compatible graph-trace state. For k particles let $\mathcal{D}_k$ be smooth compactly supported densities on nonoverlapping configurations, smooth up to non-grazing collision faces, and satisfying specular reflection and creation/annihilation compatibility. Let $T_k = v \cdot \nabla_x$ and let $\gamma_k^- g$ be the incoming kinetic trace. Set

$$\|g\|_{\mathcal{X}_k} = \|g\|_{L_w^1} + \|T_k g\|_{L_w^1} + \|\gamma_k^- g\|_{\mathcal{M}_w}, \quad \mathcal{X}_k = \overline{\mathcal{D}_k}^{\|\cdot\|_{\mathcal{X}_k}}.$$

The trace in the completion is the limit of the traces of the approximants.

Lemma 1.1 (Closed kinetic trace and Green identity). *The map $g \mapsto \gamma_k^- g$ is closed on $\mathcal{X}_k$. Every $g \in \mathcal{X}_k$ satisfies*

$$\int (T_k g) \phi + \int g (T_k^* \phi) = \int_{\partial^-} \phi d\gamma_k^- g - \int_{\partial^+} \phi d\gamma_k^+ g$$

for every smooth test function, together with the reflection and contact compatibility inherited from $\mathcal{D}_k$. Conversely an arbitrary pair of interior and boundary measures is not a state unless it satisfies these relations.

Proof. The identity holds on $\mathcal{D}_k$. Each term is continuous in the graph norm, so it passes to the completion. If $g_n \rightarrow g$, $T_k g_n \rightarrow h$, and $\gamma_k^- g_n \rightarrow \gamma$, the identity identifies $h = T_k g$ distributionally and $\gamma = \gamma_k^- g$; hence the trace graph is closed. The boundary compatibilities are linear closed relations and also pass to the limit. $\square$

Let

$$\mathcal{X}_\beta = \left\{ G = (g_k)_{k \geq 0} : \sum_{k \geq 0} \frac{\beta^k}{k!} \|g_k\|_{\mathcal{X}_k} < \infty \right\}.$$

Theorem 1.2 (Positive boundary-renewal semigroup). *The free transport with specular reflection and the creation operators $B_{k,k+1}$ define a closed dissipative generator on $\mathcal{X}_\beta$ and a positive strongly continuous semigroup. Collision-time Dirac traces belong to its graph closure, and*

$$\|B_{k,k+1} g_{k+1}\|_{\mathcal{X}_k} \leq C(k+1) \|g_{k+1}\|_{\mathcal{X}_{k+1}}.$$

Proof. Solve the resolvent along backward characteristics until the first boundary hit. The reflection map is flux preserving and the contact creation kernel is a positive boundary integral. For sufficiently large resolvent parameter the boundary renewal is a contraction; its Neumann series gives the resolvent and preserves positivity. The flux measure of one selected partner contributes at most C and there are $k+1$ partners, proving the bound. Factorial weights make the triangular hierarchy operator relatively bounded. Lumer–Phillips then gives the semigroup. Approximating a collision trace by incoming thin layers shows that its Dirac limit belongs to the graph closure. $\square$

1.2. Monotone Jacobi geometry from the youngest fork. For an orthogonal Jacobi field write (J, V) for position and velocity variations and

$$\mathcal{Q}(J, V) = \langle J, V \rangle.$$

The unstable cone is $\mathcal{Q} \geq 0$ with positive collision curvature. A free flight of length t sends (J, V) to $(J + tV, V)$. At a dispersing collision the curvature operator gains a positive term $2\mathcal{K}/|v \cdot n|$.

Lemma 1.3 (Dispersing monotonicity). *Along a non-grazing hard-sphere collision path, the unstable Jacobi norm*

$$\|(J, V)\|_*^2 = |V|^2 + \mathcal{Q}(J, V) + |J|^2$$

does not lose a transverse exterior two-volume by more than a factor $C\delta^C$ depending only on the minimal angle and event gap δ , not on the number of later collisions.

Proof. During free flight, $\mathcal{Q}(J + tV, V) = \mathcal{Q}(J, V) + t|V|^2$. At collision, the mirror formula adds the positive quadratic form $2\langle \mathcal{K}J, J \rangle/|v \cdot n|$ to the curvature. Hence the associated positive Lagrangian graph is monotone. Apply the same calculation to the second exterior power: the Gram determinant of two unstable fields is nondecreasing in the Jacobi metric. On a finite-horizon chart with the final free-flight gap at least δ , the Riccati formula $B(t) = B_0(I + tB_0)^{-1}$ gives $0 \leq B(t) \leq \delta^{-1}I$ before the final impact. Thus the Jacobi norm and the physical position exterior norm are equivalent with a fixed $C\delta^{-C}$ factor. Conversion to impact coordinates uses only the final non-grazing collision chart and is independent of the number of earlier collisions. $\square$

At the youngest separating fork of two labels choose the two root variations consisting of transverse relative translations of the two child clusters. They form an unstable two-plane immediately after the fork.

Theorem 1.4 (Depth-independent first-surplus Gramian). *On the good chronological set where all speeds, angles, and event gaps exceed δ , the derivative from the two fork variables to the relative contact coordinate at the first surplus collision has Gramian*

$$\lambda_{\min} \mathcal{G}_{ab} \geq c\delta^C,$$

with C independent of ancestral depth. The complementary set has flux measure at most $C^m \delta^\alpha$ in a depth- m sector.

Proof. At the fork the two relative translations span the contact transverse plane with a uniform determinant. They lie in the unstable Jacobi cone. Subsequent hard-sphere collisions and free flights preserve their exterior volume by Lemma 1.3; later dispersing impacts therefore cannot focus them to zero. The final impact-coordinate conversion costs only δ^{-C} . The bad-set estimate follows by integrating one speed, angle, or event-gap variable at the first failing event; the number of possible locations is exponential in m but is later absorbed by the hierarchy factorial. $\square$

Corollary 1.5 (Summable recollision sectors). *The cyclic depth- m contribution is bounded by*

$$\frac{(CT)^m}{m!}(\varepsilon^2\delta^{-C} + \delta^\alpha).$$

Choosing $\delta = \varepsilon^{2/(C+\alpha)}$ and summing over m gives $CT\varepsilon^\eta$ for some $\eta > 0$.

Proof. Coarea at the surplus contact gives the first term using Theorem 1.4; the bad chronological set gives the second. The exponential genealogy count is absorbed by $m!^{-1}$. Optimize in δ and sum the exponential series. $\square$

1.3. Source-uniform one-block convergence. Let $\mathcal{S}_{h,\varepsilon}^H$ be the exact microscopic graph semigroup with a bounded contact source H and let $\mathcal{S}_h^H$ be the Boltzmann graph semigroup.

Theorem 1.6 (One-block graph-domain estimate). *For every reached graph radius and source ball,*

$$\|\mathcal{S}_{h,\varepsilon}^H - \mathcal{S}_h^H\|_{\mathcal{X}_\beta \rightarrow \mathcal{X}_{\beta'}} \leq C_R(h\varepsilon^\eta + \varepsilon^\eta),$$

and the same bound holds for every fixed number of source derivatives.

Proof. Expand both semigroups by the boundary-renewal Duhamel series. Noncyclic trees are matched by the Boltzmann–Grad collision change of variables in the graph trace coordinates. Cyclic trees are bounded by Corollary 1.5. Differentiating the bounded source only marks collision vertices and is absorbed by a smaller factorial radius. The remainder from two events inside one block is $O(h\varepsilon^\eta)$ by the same coarea estimate. $\square$

Theorem 1.7 (Stable source sewing). *Products of one-block operators converge, uniformly over partitions whose mesh tends to zero, to one nonlinear pressure semigroup. The limit is independent of the partition and retains the actual reflected future after every contact.*

Proof. The graph semigroups are uniformly Lipschitz on reached source balls. A standard telescoping sum uses Theorem 1.6; consistency of adjacent blocks follows from the exact Markov semigroup law in Theorem 1.2. The state is never stopped or deleted at a surplus contact, so the reflected future remains in every factor. $\square$

1.4. Balance-preserving regularization. Let $\mathfrak{G}_h$ be a heat kernel on the compact group generated by spatial translations and small canonical velocity transformations preserving total momentum, energy, and the hard-sphere collision manifold. Average an entire microscopic path law under this group and then take its density/contact coordinates.

Lemma 1.8 (Commuting path-law smoothing). *The regularization R_h obtained in this way preserves positivity, total mass, momentum, energy, and the exact weak density–contact balance. It maps finite action states to smooth compatible graph states and $R_h(f, \Gamma) \rightarrow (f, \Gamma)$ in the action topology.*

Proof. Every group element conjugates the hard-sphere flow and its contact current, so pushforward commutes with the microscopic continuity identity. Averaging therefore preserves the identity exactly. Heat-kernel convolution on the group smooths every noninvariant direction. Convexity of relative entropy gives action contraction, and approximate-identity convergence gives the limit. $\square$

Theorem 1.9 (Grand-canonical and microcanonical dynamic LDP). *The sewn pressure has the full good convex dual on compatible density–contact paths. Smooth positive exposed paths give the lower bound by source tilting; Lemma 1.8 extends it to every finite-action path. B1 coefficient transfer then yields the microcanonical LDP in the order $GC \rightarrow B1 \rightarrow MC$.*

Proof. The upper bound follows from exponential tightness and the source Laplace principle. On a smooth compatible path, finite projected exposing equations have uniformly positive covariance on the quotient of collision invariants; solve them and pass through the source sewing theorem. Approximate a general finite-action path by R_h and use lower semicontinuity plus entropy contraction. Finally apply the exact shell ratio of B1 to every finite source projection and then use exponential tightness. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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